Lesson 6
Bandpass digital transmission

1 Problemas básicos

Este primer bloque de problemas son problemas extraídos de la bibliografía de la asignatura, y consisten en algunos cálculos básicos que es necesario dominar.

Problem 6.1

[Sklar2001] Calculate the expected value of the number of error bits during a day for the coherent BPSK receiver described below, under continuous operation. The data rate is $5000 bits/s$. The input digital signals are $s_1(t) = A \cdot cos(\omega_p t)$ and $s_2(t) = -A \cdot cos(\omega_p t)$, where $A = 1mV$, and the unilateral noise power spectral density is $N_0 = 10^{-11} W/Hz$.

Results for problem

2338 bits

Problem 6.2

[Sklar2001] A coherent BPSK system operating continuously produces errors at an average rate of 100 errors per day. The data rate is $1000 bits/s$. The unilateral noise power spectral density is $N_0 = 10^{-10} W/Hz$.

a) If the system is ergodic, which is the average error probability?

b) If the average received power is adjusted to 10^{-6} W, would this value be enough to keep the error probability calculated in a)?

Results for problem

a) $1.16 \cdot 10^{-6}$
b) No

Problem 6.3

[Haykin2001] The signal component of a coherent PSK system is defined by the expression

$$s(t) = A_c k sen(\omega_p t) \pm A_c \sqrt{1-k^2} cos(\omega_p t)$$

where $0 \leq t < T_b$, and the plus sign corresponds to the 1 symbol, and the minus sign corresponds to the 0 one. The first term on the right hand side of the equation represents a carrier component, included to improve the synchronization between transmitter and receiver. Solve this:

a) Plot the constellation of the signals described; what can be said about this diagram?

b) Show that, in presence of zero-mean additive white Gaussian noise with power spectral density $N_0/2$, the average error probability is

$$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}(1-k^2)}\right)$$

with $E_b = \frac{1}{2}A_c^2 T_b$

c) Assume that 10% of the transmitted power is located in the carrier component. Determine the value of E_b/N_0 required to obtain an error probability of 10^{-4}.

d) Compare this E_b/N_0 value with the one required in a conventional PSK system with the same error probability.

Results for problem

a) PSK constellation
b) Demonstration
c) $\frac{E_b}{N_0} = 8.02$
d) $\frac{E_b}{N_0} = 7.22$

Problem 6.4

[Haykin2001] We want to compare two data transmission bandpass systems. One of them employs 16-PSK, the other, 16-QAM. Both systems have to provide an average symbol error probability of 10^{-3}. Compare the signal-to-noise requirements of said systems.

Results for problem

$\Delta\left(\frac{E_s}{N_0}\right) = 3.68dB$

Problem 6.5

[Sklar2001] If the performance criterion of a system is the bit error probability, which one of the following modulation schemes would be chosen to operate in an AWGN channel? Show the calculations.

a) Non-coherent binary orthogonal FSK with $E_b/N_0 = 13dB$.

b) Coherent binary PSK with $E_b/N_0 = 8dB$.

Results for problem

Non-coherent binary orthogonal FSK

Problem 6.6

[Haykin2001] A system transmits binary data at a rate of $2.5 \cdot 10^6$ bps. During transmission, zero-mean additive white Gaussian noise with power spectral density $N_0/2 = 10^{-20}$ W/Hz is added to the signal. In the absence of noise, the amplitude of the received sinusoidal signal for digit 1 or 0 is $1\,\mu V$. Determine the average symbol error probability for the following system configurations:

a) Coherent binary FSK.

b) Coherent MSK.

c) Non-coherent binary FSK.

Results for problem

a) $9,67 \cdot 10^{-4}$
b) $5,41 \cdot 10^{-6}$
c) $3,36 \cdot 10^{-3}$

Problem 6.7

(May 2016 exam) A digital communications system uses an FSK modulation scheme to transmit two orthogonal and equiprobable signals, $s_1(t)$ and $s_2(t)$, at a rate of $2.5 \cdot 10^6$ bps. The system is characterized by $E_b/N_0 = 12$ dB.

$$s_i(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos(\omega_i t), & 0 \leq t < T_b \\ 0, & \text{otherwise} \end{cases}, \quad i = 1, 2.$$

a) Calculate the average error probability for FSK modulation schemes with coherent and non-coherent receivers.

b) Which of the previous schemes is more suitable for a channel modeled with AWGN noise? Justify your answer. For the rest of the exercise, consider only the scheme selected in this section.

c) Calculate the minimum bandwidth of the scheme selected in part b).

d) For that scheme, determine an orthonormal basis that allows representing $s_1(t)$ and $s_2(t)$. Plot the constellation in signal space indicating the coordinates of the signal vectors.

e) If you compare the previous modulation with a coherent BPSK with the same P_e, which one will require a higher E_b/N_0 ratio? Compute the increase in E_b/N_0 expressed in dB.

Results for problem

a) $P_e^{coh} = 3.43 \cdot 10^{-5}$, $P_e^{no-coh} = 1.81 \cdot 10^{-4}$
b) FSK with coherent receiver
c) $B_{min} = 6.25$ MHz

d) Basis: normalized $s_1(t)$ and $s_2(t)$.

e) FSK. E_b/N_0 increase of 3 dB.

References

[Haykin2001] Simon Haykin. Communication Systems, 4th Ed. John Wiley and Sons, 2001.

[Sklar2001] Bernard Sklar. Digital Communications, 2nd Ed. Prentice Hall, 2001.